

Retrieve factors and order of operations

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the HCF and LCM of 24 and 36. Copy or complete the first step.

2. Calculate $100 - 6 \times (8 + 3)$.

3. Is 221 prime?

4. Miro says: Miro assumes brackets always make an expression larger. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the HCF and LCM of 24 and 36. Copy or complete the first step.
Answer HCF 12; LCM 72.
2. Calculate $100 - 6 \times (8 + 3)$.
Answer 34.
3. Is 221 prime?
Answer No, $221 = 13 \times 17$.
4. Miro says: Miro assumes brackets always make an expression larger. Is this correct? Explain.
Answer Brackets change the order. The result may increase, decrease or stay the same.

Retrieve factors and order of operations

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $100 - 6 \times (8 + 3)$.

2. Is 221 prime?

3. Insert brackets and operations between 8, 6, 4 and 2 to make 40.

4. Identify and correct this misconception: Miro assumes brackets always make an expression larger.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Calculate $100 - 6 \times (8 + 3)$.

Answer 34.

2. Is 221 prime?

Answer No, $221 = 13 \times 17$.

3. Insert brackets and operations between 8, 6, 4 and 2 to make 40.

Answer $8 \times 6 - 4 \times 2 = 40$. Other correct expressions are acceptable.

4. Identify and correct this misconception: Miro assumes brackets always make an expression larger.

Answer Brackets change the order. The result may increase, decrease or stay the same.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Retrieve factors and order of operations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Insert brackets and operations between 8, 6, 4 and 2 to make 40.

2. Explain why this reasoning fails: Miro assumes brackets always make an expression larger.

3. Create a new example that tests the same idea as: Is 221 prime?

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Insert brackets and operations between 8, 6, 4 and 2 to make 40.
Answer $8 \times 6 - 4 \times 2 = 40$. Other correct expressions are acceptable.
2. Explain why this reasoning fails: Miro assumes brackets always make an expression larger.
Answer Brackets change the order. The result may increase, decrease or stay the same.
3. Create a new example that tests the same idea as: Is 221 prime?
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.